

Florian Perret

Engineering Student at Ensimag - Seeking Critical Systems Internship Abroad

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As a second-year engineering student at Ensimag, I recently completed a research internship bridging synchronous programming with parallel multi-core execution on a Real-Time Operating System. I am actively seeking an internship abroad in the critical embedded systems industry to leverage my low-level architecture background. While particularly interested in expanding my expertise in formal methods, worst-case execution time, and multi-core interference analysis, I remain highly open to exploring other challenging domains within safety-critical environments.

Education

- 2026 – Present **Engineering Degree in Computer Science and Applied Mathematics**, *Grenoble INP - Ensimag*, Grenoble
Currently in the 2nd year (Master's level equivalent). Focus on advanced software engineering, critical systems and FPGA design.
- 2023 – 2026 **Bachelor's Degree in Computer Science**, *Université Grenoble Alpes*, Grenoble
Graduated with **Honors**. Coursework included Advanced Algorithms, Software and Hardware Architectures.
- 2022 – 2023 **Preparatory Classes for Grandes Écoles (CPGE)**, *Centre International de Valbonne*, Valbonne
MP2I track (Maths, Physics, Engineering, Computer Science). Developed capacity for workload and scientific rigor.
- 2022 **High School Diploma (Baccalauréat)**, *Lycée Les Catalins*, Montélimar
Graduated with **Highest Honors**. Specialization in Mathematics and Physics-Chemistry.

Professional Experience & Research

- June 2026 – August 2026 **Research Intern in Critical Embedded Systems (Click for Poster)**, *Verimag Laboratory*, Grenoble
 - **Multi-Core Generation Framework**: Extended a framework generating parallel C code from sequential Lustre code to support deployment on Trampoline RTOS.
 - **Multi-core Interference Analysis (MIA)**: The framework provided a realistic execution environment for testing MIA tools, leading me to study interference analysis for critical systems.
 - **Low-Level RTOS Porting**: Ported Trampoline RTOS to the Raspberry Pi Pico by implementing the multi-core boot sequence and hardware synchronization to support the testing environment.
- Since 2022 **Freelance Web Developer**, *"Le Moulin de Bruno" Bakery*
Designed, developed, and maintained a showcase website (HTML/CSS/JS) with SEO optimization.

Academic & Technical Projects

- 2025 **Linker Development**, *Bachelor Project*
 - Implementation of the **merging** phase for **ELF** object files on the **ARM** architecture in C.
 - Handled symbol resolution, section concatenation, and **relocation** calculations.
 - Low-level manipulation of binary files and strict compliance with technical specifications.
- 2024 **UNIX Mini Shell**, *Bachelor Project*
 - Built a UNIX command-line interpreter in C that manages processes, pipelines, and input/output redirection.
 - **OS Concepts**: Implemented job control with foreground/background execution, utilizing core system calls (fork, exec, waitpid) and signal handling (SIGCHLD, SIGTSTP).
- 2024 **Distributed FTP Server**, *Bachelor Project*
 - Developed a C-based client-server architecture featuring a load-balancing master server and multiple slave nodes for distributed file storage.
 - **Networking**: Managed TCP/IP sockets, Round-Robin request routing, and remote file operations (GET, PUT, RM, LS).

Technical Skills

- Programming C, C++, OCaml, Java, C#, Python, SQL, ARM Assembly, Lustre, OIL.
- Systems Trampoline RTOS (OSEK/VDX, AUTOSAR), Linux.
- Tools Git/GitLab, VS Code, CMake, LaTeX, Unity.

Languages & Interests

- French Native
- English **C1 Level (Advanced)**
- Interests Embedded Systems, Cybersecurity, Edge AI.